A Machine Reads Grothendieck's Working Notes: Transcribing and Encoding Mathematical Manuscripts

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Abstract

This article reports the transcription by a large language model of the working notes of the mathematician Alexandre Grothendieck (1928–2014), held by the University of Montpellier: 178 openly accessible folders, 16,074 digitised pages that the inventory dates from 1953 to 1991, in a fast hand and a notation of his own. Twenty folders are already edited by others. The model transcribed the other 158, twenty pages at a time, into a restricted LaTeX with nine apparatus macros. A converter derives from each file a TEI document in which the mathematics is carried as TeX inside `<formula>` and commutative diagrams as `tikz-cd` inside `<figure>`. A second register, a modernised reading in current notation, is derived from the transcription and never from the hand. The transcriptions, their TEI and their PDFs are published beside Montpellier's own facsimiles, which the edition streams through a relay and never stores. In all, 638 passes have read 9,414 pages in 164 folders, and the apparatus records 63,178 words left illegible, 6.4 per cent of the words met. The article sets out the encoding and its reasons, the registers of names, dates, letters and hands derived from the transcription, and the limits: no batch has been checked by a person, and only individual points have been read against the facsimile; one macro carried two senses; and TEI's formula element carries mathematics without describing it. The use of generative AI is declared.

Keywords. mathematical manuscripts; large language models; transcription; critical apparatus; mathematical notation; archival rights; Alexandre Grothendieck.

1 Introduction

Alexandre Grothendieck handed his mathematical papers to his former student Jean Malgoire in two parts, in the summer of 1990 and on 28 July 1995. Malgoire kept them until 2010, deposited them at the University of Montpellier, and gave them to it in 2012 (Université de Montpellier n.d.). The university catalogued the fonds in 2015 and 2016 and has digitised part of it: 178 folders in open access, 16,074 pages by the inventory's count, in twenty-two groups in the order he filed them. They are working notes, written for himself in ink and pencil, over and between his own lines. He took the same chapter up four times, abandoned a proof and resumed it thirty pages later, and wrote on the backs of administrative letters, seminar notes and other people's typescripts.

Mathematicians have transcribed some of it by hand, over years. Their work covers twenty folders and 5,129 pages, a third of the open pages, in five works. *Les Dérivateurs* is transcribed whole into LaTeX (Künzer, Malgoire and Maltsiniotis n.d.; five folders, 2,135 pages). Four

works are edited in part: the *Esquisse d'un programme*, whose published text covers the two copies in its folder, pages 2–59 and 78–135 (Grothendieck 1997; one folder, 313 pages); the first thirty-seven sections of the *Longue Marche à travers la théorie de Galois* (Grothendieck 1995; four folders, 1,584 pages); the first volume of *À la poursuite des champs* (Maltsiniotis 2022; eight folders, 837 pages); and an English translation of the prenotes for EGA V by Piotr and Joseph Blass, a revised part of which Blass and Klasa published in the *Ulam Quarterly* (Blass and Blass n.d.; two folders, 260 pages). The correspondence with Serre (Colmez and Serre 2001) edits letters of which the folders hold a few. What he printed, EGA in the *Publications mathématiques de l'IHÉS*, now on Numdam (Grothendieck and Dieudonné 1960–1967), the SGA (Grothendieck et al. 1968–1977) and the FGA (Grothendieck 1962), is the text some folders draft; it does not transcribe them. The other 158 folders, 10,945 pages, had no transcription, and most of their pages had not been read since they were written.

This article reports an experiment in reading them with a large language model as the first transcriber, under constraints designed to make its reading checkable. It is the second report of a method first applied to the ledgers of Edward and Josephine Hopper (Hua 2026). It differs from that case in three ways. The document is mathematics, a notation as much as a language, so how TEI encodes mathematics becomes a central question. The corpus is some thirty times larger, 16,074 pages against the 504 sheets of the ledgers, which makes the human check the main fact about the edition. And the witness cannot be copied: the fonds is under copyright, its third-party correspondence may not be circulated without permission, and the edition shows the images only as the holder serves them.

The objective is methodological, and no argument here depends on a claim about Grothendieck's mathematics. As in the first report, a reading is *reproducible* when the process that made it can be rerun, *auditable* when a reader can find the evidence for each sentence, *reliable* when like cases are treated alike, and *valid* when it is right. The edition claims the first two properties. It measures reliability and validity only on samples, set out in Section 9.1, and claims neither for the corpus.

2 Related work

The transcription of handwriting by large language models is a recent field. Humphries and colleagues (2025) report character error rates of 5.7 to 7 per cent for general-purpose models on eighteenth- and nineteenth-century English manuscripts. Crosilla, Klic and Colavizza (2025) benchmark several models against a dedicated recognition system and find a bias towards English that they attribute to the models' pre-training. What a model has read matters most in mathematics. A model that has read much mathematics completes a formula as readily as a sentence, and since the completed formula is well-formed, nothing marks it as wrong; the model may write the index that makes an identity true where the page has another. The edition's main rule, that nothing illegible is guessed even where the mathematics would supply it, is written against this prior, and Section 9 measures its cost.

Strutz (2026) proposes a framework for evaluating model-generated TEI, from syntactic validity to semantic accuracy. As in the first report, the model here writes no TEI. It writes a restricted language whose macros correspond one to one with TEI constructs, and the TEI is derived. The precedent for a reduced language converted into conformant TEI is Cummings and Willcox (2013), whose `tei_corset` bounded what a keying company could write. Here the macros bound what a model may write, and the converter has to be testable.

The Guidelines provide `<formula>`, whose `@notation` names the notation of its content, and leave that content to another vocabulary: MathML, TeX, or an image (TEI Consortium 2026). MathML is the W3C's vocabulary for mathematical structure and presentation (W3C 2014), and a TEI file may embed it. This edition carries TeX instead, for reasons and at a cost given in Section 6. Hand attribution, which the first report treated at length, matters less here, since the fonds is in his hand, except for other people's letters and typescripts, and those are identified by what they are rather than by the letterform methods of digital palaeography (Stokes 2015).

3 The fonds

3.1 The inventory, and the three numberings

The inventory was made for an archive, and it is complete and rigorous. Each folder has a shelfmark (`19`, `140–2`, `156–9`), a title, a dating and a page count. The title is his, pencilled on the cover, or the archivists', in square brackets where the folder bore none. The folders stand in twenty-two groups, from *Cristaux* and *Théorie des motifs* to the *Longue marche à travers la théorie de Galois* and *Dérivateurs*. Each folder is served as one PDF of 4 to 270 megabytes.

Three numberings overlap on every page. Grothendieck paginated some runs himself: the Long March runs from 1 to 787 across four boxes. The archivists pencilled a number at the foot of every page, and the inventory counts these; the edition calls them pages. The university's file opens on a generated cover, so the PDF's numbering runs one ahead of the archivists'. The edition addresses a page by the archivists' number and converts to the PDF's only when it requests a page from Montpellier.

3.2 What the pages are

The pages are drafts, and a restart is transcribed like any other page. A verso is often unrelated to its recto (an administrative letter, a typescript returned by a secretary) and was digitised deliberately, sometimes upside down. Most dates are inferred. Figure 1 sets the inventory's datings against the dates the transcriptions have found on the leaves. Of the 153 folders the inventory dates, 111 rest in part on a deduction the archivists mark with square brackets, and 25 more are « s.d. », undated. The register of dated leaves holds 252 dates in 78 folders. Seventy-nine are in his hand and 85 typed. Another 55 were printed by a machine, among them the banners of the computer listings whose backs he wrote on in the 1980s; these date the paper, not the writing.

3.3 What may be read, and what was

Third-party correspondence in the fonds may not be circulated without permission. The 178 folders in open access hold 16,074 pages, or 884 batches of twenty. The twenty edited folders (5,129 pages, 266 batches) are linked to their editions and not read again, apart from a few batches in six of them that the editions do not cover. Every batch of the 158 folders that no edition covers is transcribed (Table 1).

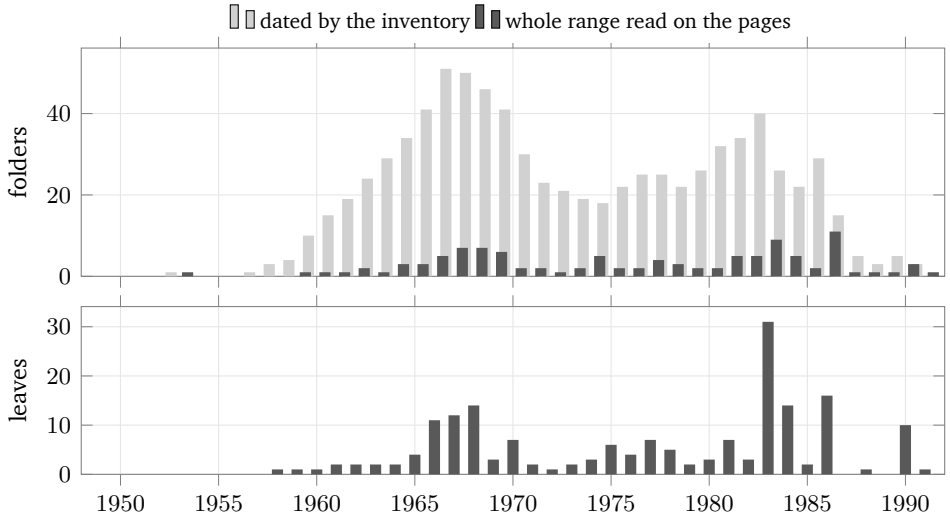


Figure 1: Two kinds of date on one axis, each on its own scale. Above: for each year, how many of the 153 dated folders the inventory's ranges cover (an open end, « à partir de 1982 », counted five years on), and, darker, how many of them rest on a range the archivists read on the pages rather than deduced in their square brackets; 25 folders are « s.d. ». Below: 182 of the 252 dates of the register of dated leaves, those written on the leaves with a year; left out are 55 printed by a machine, 9 that date the paper rather than the writing, and 6 without a year. From `src/content/catalogue.ts` and `src/content/dated-leaves.json` (2026-10-03), after the site's page *Timeline*.

Table 1: The fonds: the inventory, the part already edited by others and set aside, and what the passes have read. A batch is twenty of the archivists' pages; a transcribed page is one that carries mathematics, the others being skipped by rule.

	Folders	Pages	Batches
Open access (inventory)	178	16,074	884
Already edited, set aside	20	5,129	266
Left to this edition	158	10,945	618
Transcribed, folders left to this edition	158	9,129	618
Transcribed, edited folders	6	285	20
Read whole and modernised	164	9,414	638

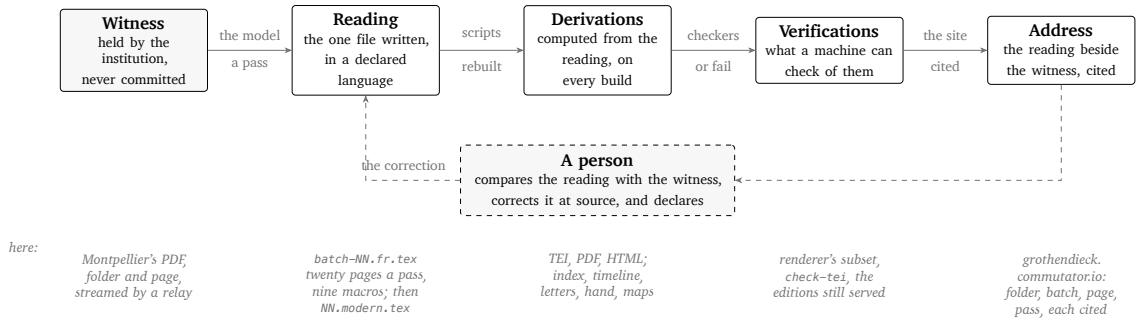


Figure 2: The schema this edition shares with its sibling on the Hopper ledgers (Hua 2026), and its values in this one. Five stations and one loop: the model turns a witness into a reading, in passes whose header names the model and the date; scripts derive everything else from the reading and regenerate it on every build; checkers fail the build on what a machine can check; the site places the reading beside the witness and cites both by reference. The dashed loop is the person, who compares the reading with the witness and corrects the reading, which alone changes anything downstream. Four invariants hold at every station: the witness is never committed; only the reading is written, everything else is derived and compared; every assertion carries an address into the witness; and the person's verdict is declared, not observed. The grey row gives this edition's values. Figure by the author.

4 The transcription method

4.1 The schema, and its values here

Figure 2 gives the arrangement described in the first report. A *witness* is held by the institution and never committed. A *reading* is the one file that is written, by the model first and by a person afterwards, in a language whose every construct is declared. *Derivations* are computed from the reading by scripts on every build. *Verifications* are whatever a machine can check of them. An *address* places the reading beside its witness and cites it by the holder's reference and the edition's dated pass, and a person closes the loop. Three things are new in this edition: a model that restates as well as transcribes, a witness that cannot be mirrored for the reader, and a notation.

4.2 The unit of work and the model

A batch is twenty of the archivists' pages, read in one fresh model context and written as a single file, `transcripts/<folder>/batch-NN.fr.tex`. The size is a design choice: a weakening of attention late in a pass would leave no mark in the output, and twenty pages bound it to one file. The skill that performs the reading admits named models only, and each file's header records the model and the date in a fixed form, `% Pass: Opus 5.5 (claude-opus-5-5), <date> --- first pass, unchecked against the pages by a human`, so that the file carries its own provenance. Of the 638 first passes, 505 were read by Claude Opus 5.5, 102 by Claude Opus 5, 26 by Claude Fable 5.1 and 5 by Claude Fable 5. The Fable models are no longer used for new passes, and the headers of the files they read say so. A hundred and seventy-five files in eighty-three folders carry a later revision below the first pass, dated and attributed in the same form. Most apply readings the editor settled on the facsimile; none makes its batch checked. Batches of one folder may run in parallel, in which case a batch does not see the one before it and its header says where a run of text crosses the boundary.

Every pass receives two aids, and neither may settle a reading. The first is a set of tiles: each page cut into six overlapping crops at 700 dpi, with the margin turned upright, so that

the pass spends no effort choosing what to look at. The second is a lexicon of his vocabulary, abbreviations and notation, counted over all transcriptions, regenerated after each run, and separating words read plainly from words read with doubt. On a hard leaf a pass usually chooses between candidates, and the lexicon shows which of them belongs to this material. Built from unverified transcriptions, it makes no claim about ink. It records that *principal* is common in these pages, and leaves to the page whether one stroke reads *principaux*.

4.3 The rules

The rule of the Hopper edition applies here with sharper consequences: an invented word in a letter is a bad reading, while an invented index in a formula can state a result nobody proved. Nothing illegible is guessed, and the mark for an illegible word is never replaced by what the mathematics would require. A doubtful reading is marked as doubtful. His spelling, punctuation, long sentences and slips are kept; slips are flagged and left uncorrected. The transcriber's notes are in French, so a French reader meets one language throughout.

The transcription is not diplomatic, and says so. It leaves aside the material apparatus of the archive: which verso carried which letter, where a separator sheet fell, which page was scanned upside down. A page with no mathematics is omitted, and the gap in the page numbers records it. Where a page is too crowded to render whole, the mathematics is transcribed and the surrounding prose compressed, with a note. Commutative diagrams go into `tikz-cd`, the notation in which this mathematics is read today. His prose stays whenever it concerns the mathematics, as when he explains why a definition is the right one. What he struck out stays, marked, because a deletion shows where the argument changed.

Three rules concern other people, and each answers a right the edition does not hold. A typescript or mémoire by someone else is given as a note per page, with his marks on it in full. A letter to him is transcribed in full where its writer is a public figure, and otherwise summarised with the writer unnamed. Letters already published in the correspondence with Serre are condensed to their statements and a summary.

4.4 What the model writes

The model writes a restricted LaTeX whose apparatus is nine macros (Table 2). The mathematics is ordinary LaTeX mathematics, inline and displayed, with the environments a working mathematician uses and `tikz-cd` for diagrams (Neves n.d.). Six declarations open each file: the shelfmark, the batch, its pages, the inventory's dating copied with its brackets, the folder's title, and a watermark, *Édition de démonstration*, which the PDF prints across every page.

Two scripts enforce the subset, the renderer of the site's reading view and the converter to TEI (Section 6), and a refusal by either stops the build. Both refuse an unknown environment; the converter further refuses a control sequence it would leave in the TEI as text, and a pass record it cannot parse. Two known faults still pass both scripts and are found only by reading the output: a backslash-newline inside inline mathematics, and a page-range declaration without a blank line before it.

4.5 A transcribed page

The following is page 8 of folder 19, *Topos*, from the first batch of that folder, as the converter exports it (the page is folder 19, batch 1). The mathematics is TeX inside `<formula>`, and the

Table 2: The apparatus macros, as the transcription skill declares them, and the TEI elements the converter maps them to. The mathematics itself is carried as TeX (Section 6).

Macro	Meaning	TEI
`\page{47}`	Page 47 begins, in the archivists' numbering	`pb` with `@n`, `@facs`
`\ill{}`	Illegible; never guessed	`gap reason="illegible"`
`\uncertain{…}`	A reading offered and flagged	`unclear`
`\supplied{…}`	What the transcriber supplies and the page does not carry: a missing letter, a clipped word-end	`supplied resp="#pass"`
`\add{…}`	His own insertion: above the line, over a struck word, his ink on a typescript	`add`
`\struck{…}`	Struck out by Grothendieck	`del`
`\note{…}`	The transcriber's note, about the mathematics	`note type="editorial"`
`\drawing{…}`	A drawing of his, described rather than redrawn	`note type="drawing"`
`\marginal{…}`	A marginal note of his	`note type="authorial" place="margin"`

diagram is a figure. The note is the transcriber's, the marginal note his, and both the struck φ and its replacement are kept.

```
<pb n="8" facs="https://grothendieck.umontpellier.fr/19.pdf#page=9"/>
<p>qui s'expliciteront en termes des <formula notation="TeX">\varphi_{ji} = f_{j} g_{i}</formula>,
     [...]
<formula notation="TeX" rend="display">\lambda_{kji} : \varphi_{ki} \longrightarrow
  \varphi_{kj} \circ \varphi_{ji} \tag{3.10}</formula></p>
<p><figure type="diagram"><formula notation="tikz-cd">\begin{tikzcd}[column sep=small, row sep=
    small]
\varphi_{ki} \arrow[r, "\lambda_{kji}"] \arrow[d, no head, "\parallel" description]
  & \varphi_{kj}\varphi_{ji} \arrow[d, no head, "\parallel" description] \\
f_{k} g_{i} & f_{k}\, g_{j} f_{j}\, g_{i}
\end{tikzcd}</formula></figure></p>
<p><note type="editorial" resp="#pass">la flèche provient du morphisme d'adjonction
<formula notation="TeX">\mathrm{id} \to g_{j} f_{j}</formula>, que l'auteur indique par une
flèche verticale sous <formula notation="TeX">g_{j} f_{j}</formula></note></p>
<p><note type="authorial" place="margin">pour <formula notation="TeX">i = j = k</formula>,
<formula notation="TeX">\lambda_{kji}</formula> n'est autre que
<formula notation="TeX">\lambda_{i}</formula></note></p>
<p>[Le cas le plus intéressant est celui où les <del><formula notation="TeX">\varphi</formula></
    del>
<add><formula notation="TeX">\alpha_{i}</formula></add> sont des
isomorphismes (i.e. les <formula notation="TeX">g_{i}</formula> pleinement fidèles). [...]</p>
```

In the last paragraph the α_i, written over the φ he struck, is his and is encoded as `<add>`; an earlier mapping credited such insertions to the transcriber, for reasons given in Section 9.3.

5 Two registers

The Hopper ledgers needed one text, and this fonds needs two because of its notation. A $\mathcal{C}$ of 1962 is not drawn like one of 1986, a sheaf of the Séminaire is not named as sheaves are named today, and a reader who meets « fppf » spelled out on one page and reduced to initials

on the next needs to be told it is the same topology. Every folder read whole therefore yields a second document, the *modernised reading*. It opens with a résumé for a reader new to the subject, then gives the mathematics in current notation and current names, with footnotes carrying everything the transcription's apparatus carried.

The modernised reading is derived from the transcription and never from the hand, since two independent readings of one hand would diverge with nothing to decide between them. It is made per folder, because his arguments do not stop where the batches do. As the one document allowed to depart from the page, it is held to another standard: it must be correct as it stands. Where the manuscript is loose, the reading states what is true and gives what the page has in a footnote. The skill turns the standard into four failure modes to check: a statement strengthened beyond what the page proves, a hypothesis dropped, a notation modernised into a different object, and a gap closed silently.

The résumé ends with a line of English keywords, the modern vocabulary to search under, and the only part of either register written for search. The site builds its search, its index of subjects and its map of the mathematics from them (Section 7.3), and labels them as the reading's. The 164 folders read whole in this way carry 3,390 keywords.

6 The TEI encoding

6.1 A derived export

A script exports every transcription to TEI P5. It reads the `.tex` and nothing else, neither the facsimile nor the rendered HTML, and its mapping is that of Table 2. The reasons are those of the first report. A model asked for TEI writes it fluently, and well-formed markup with an invented attribute shows no structural sign of the error, whereas a nine-macro language has a small surface and each of its macros answers one editorial question. Because the TEI is derived, any difference between a file and its TEI is a bug in the converter. The LaTeX is legible only to this repository; TEI can be taken in by an archive or another editor without the edition's tools.

The header carries the file's comment in machine-readable form. Its title statement declares the editor in an `<editor>` and names the model, in a responsibility statement, as a `<name type="model">`. Every editorial note points to the model, except a note that records a reading the editor checked on the facsimile, which points to him. The converter recognises such notes by a short list of phrasings that name him as the one who looked at the page; a check it misses stays with the model. The availability is `restricted`, with the reasons: the fonds is under copyright, the transcription is an unauthorised demonstration, and no image is reproduced. The manuscript description gives the shelfmark, the inventory's title, and its dating in an `<origDate>`. The surrogate is Montpellier's PDF, with a note that each `@facs` points one page ahead of its `@n`. Every pass recorded in the source is a `<change>` in `<revisionDesc>`, dated and pointing to its model, or to the editor when it records his check.

6.2 Mathematics as TeX

Mathematics is carried as TeX, unchanged, inside `<formula notation="TeX">`, with displayed formulas marked by `@rend`. A commutative diagram is a `<figure type="diagram">` holding a `<formula notation="tikz-cd">`. The few drawings that an argument depends on and that are not arrow graphs (arcs, chains of faces, pseudo-lines) are redrawn in TikZ and carried as `<figure type="picture">`. It holds the source in a `<formula notation="TikZ">` and, in

a <graphic>, an SVG compiled from it, since no browser draws TikZ. Neither TEI nor the converter encodes the mathematics, for three reasons and at one cost.

First, the transcription is already TeX, which is closer to what he wrote than MathML would be. He wrote a notation without declaring a structure: an underlined ω, a doubly underlined Σ, a tilde that distinguishes two coverings, one letter used for two objects on one page. Presentation MathML would be a second transcription of the first, and content MathML would be an interpretation, which belongs to the modernised reading. Second, an illegible exponent or a doubtful index is an apparatus mark inside a formula, written as `\ill{}` inside `$…$`. Kept as TeX, it stays where both the browser's renderer, KaTeX (KaTeX n.d.), and the PDF read it, whereas a conversion to MathML would have to give an illegible subscript a structure that nobody knows. Third, the round trip: the TeX inside the TEI is the TeX of the source, character for character, so the two can be compared as strings.

The cost is that the TEI says nothing about the mathematics that a TEI processor can use. A reader of the TEI can find every formula but cannot search inside one, and the 2,000 commutative diagrams are opaque to the schema. An apparatus mark inside a formula is invisible to TEI, so a count of <gap> elements misses every illegible index, and the edition takes its own counts (Section 7.3) from the source. MathML can be derived later from the same TeX, where the TeX is clean enough, without touching the transcription. The journal's own customisation, against which this article is encoded, admits no <formula>, so the article's TEI carries its few inline symbols as Unicode.

6.3 Validation

The edition has a customisation of its own, modelled on the one for the Hopper ledgers, where a corpus of well-formed files had turned out to be invalid because nothing had checked them (Hua 2026). Its ODD declares the sixty-five elements the converter emits, header included, and closes the value lists of the attributes that carry meaning: `@type` on , <figure> and <note>, `@notation` on <formula> (`TeX`, `tikz-cd`, `TikZ`), `@reason` on <gap>, and `@rend` wherever the converter writes it. As a pure subset, it accepts only files that are valid TEI.

The RELAX NG schema is derived from the ODD on every build and published beside it, and every exported file names it in an `xml-model` processing instruction. The build validates every file with jing (Clark 2022) against unmodified `tei_all` and against the derived schema, with every input pinned by version and digest. It compares the ODD with the export in both directions and fails on an element used but not declared, or declared but never used. A further check typesets every formula of the TEI's reading view and of the `.tex`'s with KaTeX and compares them page by page. Two things are not validated: the Schematron rules that `tei_all` embeds, which are not run over the corpus, and the TeX inside each <formula>, which the schema sees as a string and only KaTeX reads.

6.4 Several witnesses of one text

Some texts survive in more than one copy, each with its own marks. The *Tapis de Quillen*, sixteen typed pages dated 10 September 1968, has four witnesses. In folder `162-5` of the Montpellier fonds, the typed top copy carries his corrections in blue ink, and a black photocopy taken after the ink adds marks of its own; folder 111 is another photocopy of that second state. A fourth copy, in the Fonds Grothendieck–Illusie of the IHÉS (item 2ILL3; IHÉS n.d.), was photocopied from the inked top copy before those later marks, and carries pencil notes

written on it alone. The letter to Tate of May 1966 is in Montpellier folder 6 and, as a photocopy of the same annotated sheets taken before Montpellier's numbering, in the IHÉS item FG_IL_2_TATE66. Pages 168–185 of folder 93, on the Cartier operation, are also IHÉS item FG_IL_3_2, which keeps a corner that the Montpellier sheet has lost.

The copies were identified by internal evidence: the same typed master, with the same line breaks and the same typewriter x-outs, and a page-to-view correspondence that holds through the whole item (view N of the Tate letter is page $N+1$ of folder 6). Each witness is transcribed in its own file, against its own images. When another witness settles a reading, the correction is made in the file it concerns, and its header records a revision that names the witness and the view it was read on. A note in the body says which witnesses carry the mark. In this way the edition corrected some thirty readings in folders 6, 93 and 111 from the IHÉS copies, among them two marginal blocks of the *Tapis* that folder 111 had given as unreadable.

TEI allows more. A parallel-segmentation apparatus would declare the witnesses once in a `<listWit>` and record each variant where it occurs, as in this schematic case from leaf 3 of the *Tapis*, where a marginal "Si !" with its bracket is on three witnesses and absent from the top copy:

```
<listWit>
  <witness xml:id="B">Montpellier 162–5, typed top copy, blue ink</witness>
  <witness xml:id="A">Montpellier 162–5, black photocopy with its own marks</witness>
  <witness xml:id="M">Montpellier 111, photocopy of A's state</witness>
  <witness xml:id="I">IHÉS 2ILL3, photocopy of the inked top copy</witness>
</listWit>
[...]
<app>
  <rdg wit="#A #M #I"><note place="margin">Si !</note></rdg>
  <rdg wit="#B"/>
</app>
```

The edition does not do this yet. Each exported file encodes one witness, and the agreement between witnesses lives in prose notes, which a TEI processor cannot query. Such an apparatus would further need the mathematics to be segmented, since a variant often falls inside a formula, where the TeX string has no internal structure (Section 6.2).

7 Interface and reproducibility

7.1 The facsimile

The site at `https://grothendieck.commutator.io` shows the transcription in one pane and Montpellier's facsimile in the other. Scrolling the transcription past a page break turns the facsimile to that page. A reading is addressed by page: `#140-1/2/p28` opens folder `140-1` at batch 2, scrolled to page 28, with the facsimile at the same page. Every register below links in that form. The edition serves no image of the fonds: the repository's ignore file refuses PDF files, and the server holds none.

A frame pointed at Montpellier's server fails, for three reasons. The server sends the header `X-Frame-Options: SAMEORIGIN`; its certificate expired on 10 December 2025, and it sends no cross-origin headers. The browser enforces all three against the remote origin, but none applies to a request made by a server, so a relay on the edition's origin requests the file and passes the bytes through unchanged, with range requests, so that opening page 221 of a 204-megabyte folder transfers 256 kilobytes. The relay serves only an allowlist of the folders' files,

checked against the inventory, and stores nothing.

The reading can be checked against its witness only while Montpellier's server is online. When the server is down the pane says so and offers no copy; the local mirror of page images that the passes read from serves the transcription only and is neither served nor committed.

7.2 Counts computed from the files

Everything the site shows beyond the transcription is derived from it, and every count, including the manifest of what is transcribed, is computed from the files present. A batch counts as transcribed once its LaTeX exists, and as modernised once the folder's reading exists; neither status claims that anyone has checked it. The site's progress figure is computed against the folders nobody has edited, and the figure against the whole fonds is printed beside it.

7.3 Registers derived from the reading

Because the transcription is written in a constrained language, a question put to the fonds can be answered by a script. The edition publishes seven such registers, each regenerated from the files, with every entry resting on a file and a line.

The hand counts the apparatus per batch and per folder. Of the words met, 925,227 were read and 63,178 left illegible, 6.4 per cent; 44,516 readings were marked doubtful. He struck 39,341 times and inserted 14,757 times outside the transcriber's notes; the transcriber supplied 720 letters or words; there are 5,796 marginal notes, 2,000 commutative diagrams and 902 drawings, four of them redrawn and the others described in notes. Figure 3 places each folder's share of unread words against its inventory dating. What is counted is the apparatus: an `\ill{}` is a word one pass could not read, so the count depends on the pass as well as on the hand.

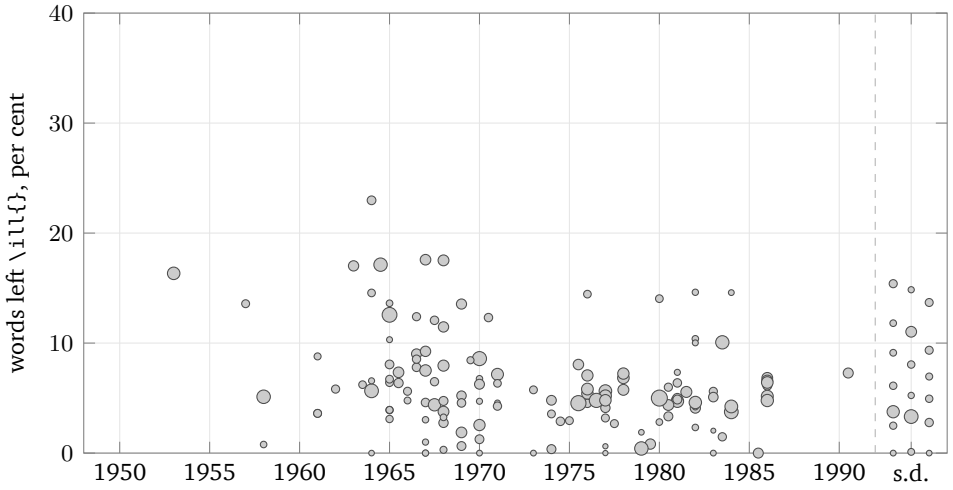


Figure 3: The share of words the first pass could not read, folder by folder, against the middle of the inventory's dating; 19 undated folders stand in the column at the right. A larger dot, more pages transcribed. 164 folders, 925,227 words read and 63,178 left illegible (6.4 per cent). No trend is drawn: the datings are ranges, often decades wide. The count is of the apparatus, not of the ink. From `src/content/hand.json` (2026-10-24), after the site's page *The hand*.

The timeline draws each folder's inventory dating as a range, dark where the archivists read it and light where they deduced it, with the dates on the leaves as dots (Figure 1 summarises it). The parser keeps the qualifier of each end of a dating, « vers », « à partir de », « avant »;

reducing « 1965-[vers 1971] » to 1965–1971 would lose the distinction the page records. *The letters* is a register of 93 letters in 41 folders, 1959 to 1991, each with its date as written, its direction, its form, and how the transcription gives it, private correspondents unnamed. *The index* leads from a person, a work of his cited by volume (EGA, SGA, FGA), or a subject, to the folder and batch where it occurs. *The maps* are co-occurrence networks in the manner of bibliometric mapping: who is named together in a folder, which subjects share folders, which folders cite which volumes. Every edge rests on evidence, the layout uses a fixed seed, and a slider shows each map with only the folders dated by a given year, by the midpoint of the inventory's dating.

The last two registers collect the mathematics itself. *The diagrams* are the 2,000 `tikzcd` environments of the transcriptions, in 142 folders, shown one to a slide by the reading view's renderer. *The formulas* are the 25,946 displayed formulas, in 156 folders, set by KaTeX with their apparatus. Both open on their richest items, ranked by a stated count: the distinct symbols a diagram or formula uses (a letter in a given alphabet, so that $\mathcal{C}$ is not C, a named symbol, an operator), plus nodes and arrows for a diagram and size for a formula. The ranking measures quantity, not importance. The richest diagram has nineteen objects and twenty-nine arrows, on page 28 of folder `140–1`, a draft of the *Longue Marche*.

Two rules govern these registers. The transcriber's notes are excluded from every count of his words, since a note discussing a notion would otherwise report it as his. A person counts as a node where the folder holds a letter to or from them, a text of theirs, or his naming them as the source of a result, as in « th. de Hodge » or « d'après Deligne ». A mathematical object that bears a name, such as « groupe de Galois », does not count; the test is adjacency in the text. Sources are held to a further condition: one judged unreliable is not used, even where it is the only source for a claim, which then stays unsourced and is left out of the graph.

8 Findings

Reading a folder end to end turns up things the inventory does not record: a leaf bound backwards, two manuscripts interleaved, a theorem proved under a hypothesis weaker than the one in print. The edition keeps them in a list designed to prevent overclaiming. To say that something in these folders is new is to make a claim about the literature, which can be searched but never exhausted, and to say that it came first needs a date, which the folders mostly lack.

Each row carries a status from a closed list: *unsearched*, nobody has looked it up; *candidate*, searched and not found; *matched*, found in the literature and kept; *refuted*, shown false as stated by a counterexample, kept with what still stands; *confirmed*, which only a person may set. Every row names the sources searched and separates what the page carries from what the edition supplied. The skill that proposes rows reads the transcription and the modernised reading, never the facsimile, and may not assert priority. Matched and refuted rows stay in the list, which spares the next reader the search. The list holds 463 rows from 155 folders: 216 unsearched, 165 candidates, 73 matched, eight refuted, and one confirmed, on the word of the author of the book it corrects.

One row shows the purpose of the list. Folder 27 holds five typed leaves headed in pencil « Exp XIII »: a proposition on a group acting on a scheme, with a chain of implications among six conditions, and three corollaries. A first pass summarised them as an already published typescript, but a comparison with the exposé in the current re-edition of SGA 3 (Gille and Polo

2008–2011, Exposé XIII in its version of December 2008) showed the premise was wrong. The published exposé goes from its first lemma directly to its second section, and the general statements reappear there only in specialised form. The leaves are an earlier, longer version that is not in print, and they were re-transcribed in full. The comparison shows that the inventory's « typescript » and the published exposé are different texts, which a reader of the folder could not have known; it does not show that the earlier version matters.

Statements of the readings of seventy-two folders are proved in Lean 4 (de Moura and Ullrich 2021) against mathlib (mathlib Community 2020), in 2,119 theorems, counted as the declarations that open a line with `theorem`, all resting on Lean's standard axioms only. A proof shows that the reading's statement holds, which does not make it what the page says; in some folders it found a statement false as given.

9 Discussion

9.1 The reading against the image

The transcription is a first pass that no person has checked batch by batch. The editor has checked individual points against the facsimile: dates, doubtful words, the hand of an addition, the ink of typed batches sampled one page each, untranscribed pages (found blank), the sixty-three points at which eight documents differ from a transcription made independently elsewhere, and a few pages of letters and typescripts. A hundred and sixteen batch files record such checks, each dated and naming the editor as the reader of the facsimile; the page of folder 19 in Section 4.5 was read in this way, and four of its readings were corrected (Appendix A). None of these checks makes a batch checked. Only a person may declare a batch checked or skipped, in the edition's status file, which declares two skipped and none checked, and the site shows every batch as drafted. The apparatus states the rate at which the reading fails: 6.4 per cent of words left illegible and a further 4.5 readings in a hundred words marked doubtful, the share left illegible varying by folder from none to 23 per cent (Figure 3). These are not error rates, since a word misread plainly leaves no mark and only a person with the facsimile can find it. Two small samples, neither drawn at random, come closer to one. At the sixty-three points where eight documents differ from the other transcription, the facsimile kept the machine's reading at forty-four and changed it at nineteen, in six of the eight documents. Of the twenty-seven illegible words the review page sampled, the editor read twenty on the facsimile and seven stayed illegible; of nineteen doubtful readings, fourteen were settled, four stayed illegible and one stayed open.

9.2 Checking by spot: the review page

Many of these checks were made through a page of the site, `/review/`, which turns the edition's checklist into single questions. The checklist, first the body of issue #35 in the repository, is kept as a file, `docs/review/issue-35.md`. A script, `scripts/review-queue.mjs`, reads it with the transcriptions and builds the queue: in the batches densest in unread words it samples three `\ill{}` and two `\uncertain{}` per batch; over the corpus it samples free-standing `\add{}` and glued `\supplied{}`, one folder at a time; and it adds the dated leaves and the register of letters whose entries the checklist questions. Each item is one spot of one page, shown beside Montpellier's facsimile, opened at that page, with a question in plain words (« Is the highlighted word what the page has? ») and a fixed set of answers, one of which asks for the

reader's own reading in at most fifteen words.

Answers stay in the reader's browser until they are sent. Sending opens a GitHub issue already filled in, which the reader submits under their own account, and the answers are then applied at source. The batch file gains a line `% Revised:` naming the model that applied the change, the date, the reader and the issue, and `scripts/tei.mjs` turns each such line into a `<change>` element, with `@when` and `@who`, in the `<revisionDesc>` of that batch's TEI header. Where a reading needs it, a `\note{}` on the spot says what was checked. The item is recorded in `review-answered.json` with its issue and a result (applied, confirmed, still illegible, still open), and the next build of the queue leaves it out.

Whether the site may show the transcriptions is not yet settled with the holders of the rights. It shows them for now, and a switch read at build time, `RENDER_TRANSCRIPTS`, turns their display off in one setting. Because the issues are public, they carry only the verdict and the reader's fifteen words. The current reading around each spot, likewise at most fifteen words, sits in a separate file that the page imports only when the build may show transcriptions; with the switch off, the file is not in the bundle.

By 23 October 2026 nineteen issues had come from the page, #61 to #65, #67 to #71, #74 to #78, #80 to #82 and #84, and settled 173 items, all answered so far by the editor. Of eighty-seven words in question, seventy-four readings were changed or confirmed, forty of them insertions whose hand was in question; eleven words stayed illegible, and two stayed open. Thirty-six items of the register of letters were confirmed or corrected. Forty-seven dates were checked on their leaves, and forty-five confirmed or corrected: the corrections place a date on its page, give it in the form the page has, mark a label or a title that is not a date of writing, or name the leaf that carries the year of a day-only date; two could not be placed. The remaining three items asked whether one of the edition's own arguments holds, or whether a reference cited from memory is right. Reading the hand does not qualify a reader to judge the mathematics, and such checks are left to specialists. The queue now holds 10 items.

9.3 Limits of the model as a transcriber and as an encoder

There are five limits, none of them an error rate.

One macro carried two senses. `\add` was first declared as the transcriber's addition (a missing letter, an implied word) and mapped to `<supplied resp="#pass">`. In use, most of its occurrences were his: a word written above the line with a caret, a substitution over a struck word, the α_i of Section 4.5. The TEI thus credited several thousand of his insertions to the model. The Hopper edition had the same fault and split the macro in two, as this edition then did. `\supplied` now carries the transcriber's sense and maps to `<supplied resp="#pass">`; `\add` carries his and maps to `<add>`. The existing uses were divided by a rule that can be checked in the file, without a new reading: an addition attached to the word it completes, with a letter or digit immediately before or after it, is the transcriber's, and a free-standing one is his. The rule assigned 529 of the 6,144 uses to `\supplied` and left 5,615 as `\add`; later passes use the two macros from the start. Its output is a guess: a closing parenthesis restored at a clipped edge still reads as his, and a prefix he inserted, now «`\supplied{quasi-}torseur`», is attributed to the transcriber. Only a person reading the facsimiles can settle these cases. A macro whose declared sense is narrower than its natural sense gets used in the natural one.

The model knows the mathematics. Its prior is the hazard the rule of Section 4.3 addresses, and the rule holds only where it is applied, since an illegible index that the context determines is easy to fill in. The transcriptions write `\ill{}` and, where the context fixes the value, say so

in a note, as in « l'indice de k est un pâté … le contexte ne le fixe pas ». A pass that supplied the index silently would leave no trace. For this reason the method separates the registers: completion is allowed in the modernised reading, which may state what is true, and forbidden in the transcription.

The apparatus is inside the mathematics. As Section 6.2 explains, a count over the TEI misses every mark inside a formula, and a user of the TEI alone would undercount the illegible words.

Other people's texts. The fonds holds letters, typescripts and mémoires by others, some of them private persons. The rules of Section 4.3 answer to rights, and the model has to apply them. The skill judges conservatively whether a writer is a public figure, and the register of letters leaves private correspondents unnamed. Only a human reviewer can check that judgement, and an error in it would disclose a private person's text.

Dates. The model reads a date as readily as a word, and the edition needs dates more than anything else, since the inventory's datings are mostly inferred (Figure 1). The register of dated leaves distinguishes a date in his hand from a typist's and a machine's, keeps two-digit years as written, and fixes a year from a neighbouring leaf only where the run is continuous, saying so. Of its 252 records, 195 were assembled by a model reading the transcriptions and have not been checked against the facsimile.

9.4 Human validation, and the pace

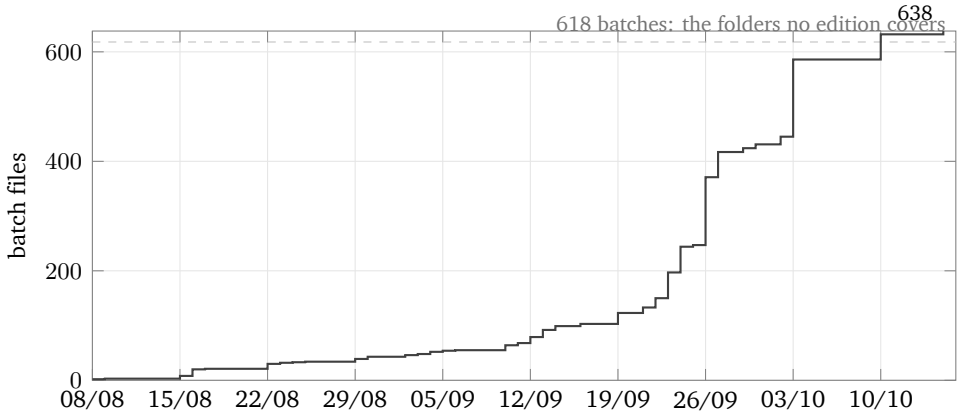


Figure 4: Batch files in the repository, day by day, from the first on 2026-08-08 to 2026-10-15: the date each `batch-NN.fr.tex` was first committed, cumulated. Each file is one pass of the model over at most twenty pages. The dashed line is the 618 batches of the folders no edition covers; the curve ends 20 above it because six edited folders also have a few batches. From the repository's git history, drawn on 2026-10-24 by `docs/study/figures.mjs`.

Figure 4 gives the pace from the repository's history. The first batch file was committed on 8 August 2026 and the 638th on 15 October: 43 in August, 388 in September and 207 in October, with up to eight passes running at once. The 618 batches of the folders no edition covers were transcribed in about eight weeks. Checking them takes far longer: a mathematician who can read this hand needs about an afternoon to compare a batch of twenty pages line by line with the facsimile, so the 618 batches represent some two and a half thousand hours of a rare kind of reader. With the transcription done, the checking is what limits the edition.

The edition proposes the protocol its status file already encodes. A person opens a batch beside its facsimile, reads each transcribed line against the page, corrects the transcription at source with a note saying who checked it, and only then declares the batch checked. Until then

every citation names a machine pass and a date, and the address of a page does not change when its reading does.

9.5 What the method permits, and what it does not

The method allows one person and a model to produce, in months, a reading of a sixteen-thousand-page mathematical archive, cited by folder, batch, page and pass, exported to TEI with its mathematics intact, placed beside the institution's facsimile without copying it, and mined for registers that each rest on a line. It does not produce a checked transcription, it does not encode the mathematics in a form a TEI processor can reason over, and it cannot decide with authority who is a public figure.

10 Conclusion

A second edition has been built on the method of the first, and a corpus of mathematics changes several things in it. The notation required a decision the ledgers did not. Mathematics is carried inside `<formula>` as the TeX it was transcribed in, without a structure the transcription does not have, and the interpretation of the notation goes into a second register derived from the first. The size of the corpus made the human check the main fact about the edition. The rights meant that the edition points to the facsimile instead of serving it, through a relay that stores nothing.

The encoding is in place and validated on every build. Two pieces of work remain. The first is to check on the facsimile the insertions that the rule of Section 9.3 divided mechanically between him and the transcriber. The second, on which every claim of the edition depends, is the human check of every batch.

For this fonds, checking means reading Grothendieck's hand against sixteen thousand pages. It needs mathematicians who can read that hand; few can, and nobody can do it alone. The method provides a cited reading before that check and does not replace it.

About the author

Michel Hua is an independent consultant at Commutator, Paris, working with large organisations on data and cloud applications: platform migrations, data governance, and the adoption of generative AI. He has built editions of archives that institutions have digitised and few have read: the Hopper ledgers at the Whitney Museum, the Grothendieck fonds at Montpellier, and mathematicians' manuscripts in Gallica. `michel@commutator.io`.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude Opus 5, Claude Opus 5.5, Claude Fable 5 and Claude Fable 5.1 (Anthropic), through Claude Code, in order to transcribe the pages of the fonds into the restricted transcription language described in Section 4, to write the modernised readings described in Section 5, to write the scripts, the converter, the relay, the interface and the derived registers described in Sections 6 and 7, to propose the findings and the Lean proofs described in Section 8, and to draft this article from the transcriptions and

the derived data under the author's direction. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication. The transcriptions have not been checked against the facsimiles by a person, and the article says so wherever a claim depends on them.

Data availability

The scripts, the converter, the relay, the interface, the derived registers, the transcriptions, the modernised readings and the sources of this article are in the repository at `https://github.com/Commutator-IO/grothendieck-project`. The interface, with the reading views, TEI and PDFs of the transcriptions, is served at `https://grothendieck.commutator.io` beside the facsimiles that Montpellier serves. The software and the derived data are released under the repository's licence, and this article under CC BY 4.0. The fonds belongs to the University of Montpellier, and the rights in the manuscripts are held by Grothendieck's heirs. The transcriptions and modernised readings are working documents for which the edition holds no authorisation; they are withdrawn at the request of the rights holders, which one build setting allows. No image of the fonds is stored by the edition or reproduced in this article; a local mirror of page images serves the transcription only and is neither served nor committed.

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A A complete transcribed page

Page 8 of folder 19, batch 1, as the model wrote it in the restricted language, in the first pass, by Claude Opus 5; the TEI of Section 4.5 is derived from the corrected file. The page is on the site, beside Montpellier’s facsimile.

```
\page{8}
qui s'expliciteront en termes des $\varphi_{ji} = f_{j} g_{i}$, et des
$\bar{\alpha}_{i}$, $\bar{\lambda}_{i}$ correspondants, ainsi
\[
  \left\{
  \begin{array}{l}
    \bar{\alpha}_{k} = \alpha_{k}(X_{k}) \circ \mathrm{pr}_{k} \\[2pt]
    \bar{\lambda}_{k,j,i} = \lambda_{kji}(X_{i}) \circ \mathrm{pr}_{i}
  \end{array}
  \right.
  \tag{3.9}
\]
\[
  \lambda_{kji} : \varphi_{ki} \longrightarrow \varphi_{kj} \circ \varphi_{ji}
  \tag{3.10}
\]

\begin{tikzcd}[column sep=small, row sep=small, nodes={font=\scriptsize}]
\varphi_{ki} \arrow[r, "\lambda_{kji}"] \arrow[d, no head] & \varphi_{kj}\varphi_{ji} \arrow[d, no
     head] \\
f_{k} g_{i} & f_{k}\, g_{j} f_{j}\, g_{i}
\end{tikzcd}

\note{la flèche provient du morphisme d'adjonction $\mathrm{id} \to g_{j} f_{j}$,
que l'auteur indique par une flèche verticale sous $g_{j} f_{j}$}

Les $(\varphi_{ji}, \lambda_{kji})$ sont reliées par des conditions exprimant
(1.5), (1.6), que nous n'expliciterons pas.

\marginal{pour $i = j = k$, $\lambda_{kji}$ n'est autre que $\lambda_{i}$}

[Le cas le plus intéressant est celui où les \struck{$\varphi$}
\add{$\alpha_{i}$} sont des isomorphismes (i.e. les $g_{i}$ pleinement
fidèles). Dans ce cas, \struck{\ill{}} on peut supposer
$\varphi_{i} = \mathrm{id}_{B_{i}}$,
$\alpha_{i} = \mathrm{id}_{\mathrm{id}_{B_{i}}}$, $\lambda_{i} = \mathrm{id}$.]
```

```
Les conditions (1.) \struck{les} dans les cas où deux des $i, j, k$ sont égaux,
avec $i = j$ ou $j = k$, \uncertain{ne montrent que} l'un des $\lambda_{kji}$
est l'identité, \ill{} de $\varphi_{kj} \varphi_{ji}$.
\note{ligne très surchargée ; le verbe reste douteux}

Il reste donc à examiner les cas (3.11) : a) \struck{$i$, $j$, $k$ deux à deux
différents} ; b) $i = k \neq j$, d'où
\[
  \mathrm{id}_{B_{i}} \longrightarrow \varphi_{ij} \circ \varphi_{ji}
  \qquad (i \neq j)
\]
```

Read against the facsimile by the editor, the page needed four corrections. The two vertical strokes of the boxed diagram (3.10) are equality signs, which the pass had drawn as plain lines. Case a) of (3.11) reads « i, j, k tous différents » and is not struck; the pass had read it as struck and as « deux à deux différents ». On a second reading (15 October 2026), the struck words after « Dans ce cas, » are « λ_{i} est », and the line on the conditions reads « deux des indices i, j, k consécutifs » and, in its right order, « dès lors que l'un des deux facteurs de φ□□φ□$_{\mathrm{i}}$ est l'identité ». The file carries all four corrections and two `Revised` lines naming them.